

UNDERGRADUATE FINAL YEAR PROJECT REPORT

Department of Computational Finance

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FORECASTING PAKISTAN PENSION SYSTEM USING MACHINE LEARNING TECHNIQUE

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Author's Declaration

We declare that we are the sole authors of this project report. It is the actual copy of the project report that was accepted by our supervisor(s) including any necessary revisions. We also grant NED University of Engineering and Technology permission to reproduce and distribute electronic or paper copies of this project report.

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Statement of Contributions

- **M. HARIS:** Performed data collection from Annual Audited Financial Statements of eight PSX-listed companies (HBL, OGDCL, PPL, SECP, NBP, SSGC, PIA, PAKOXY), data cleaning and preprocessing in Python and Excel. Built the consolidated 2015–2024 pension obligation dataset by extracting IAS 19 disclosure figures (Opening Obligation, Service Cost, Interest Cost, Actuarial Gains/Losses, Benefits Paid, Closing Obligation), structured the data into company-wise time series using Pandas, and validated figures against published financial notes to ensure dataset integrity.
- **FAISAL:** Handled data cleaning, preprocessing, and implementation of Machine Learning models and graphs using Python. Implemented and tuned five forecasting models (ARIMA, Linear Regression, Random Forest, XGBoost, and CatBoost) for each of the eight companies using scikit-learn, statsmodels, and the XGBoost/CatBoost libraries, performed hyperparameter tuning for the ensemble models, computed evaluation metrics (MSE, RMSE, MAE, R^2) for each model-company pair, and generated comparative visualizations (actual vs. predicted trends, feature importance plots) using Matplotlib and Seaborn.
- **DANIYAL GUL:** Responsible for compiling the final project report and conducting the initial literature review. Structured the overall report architecture (chapters, headings, numbering, and referencing format per APA 7th edition), consolidated model output tables and figures from the technical team into the Results and Discussion chapters, and compiled the company-specific recommendations in Chapter 5 based on the comparative model performance findings.
- **LAIQA:** Conducted the literature review, synthesizing 35+ national and international studies on pension forecasting and ML applications into the comparative framework used to guide model selection in Chapter 3. Standardized the visualization template (color schemes, axis formatting, legends, and layout) applied consistently across all model-company result charts to ensure clarity and uniformity, and performed full referencing and formatting quality assurance across all chapters, ensuring APA 7th edition compliance throughout the report.
- All group members jointly participated in research design, results interpretation, and final proofreading of the project report.

Executive Summary

Pakistan's pension system operates under a Pay-As-You-Go (PAYG) model that is increasingly strained by a growing elderly population, rising life expectancy, and limited fiscal capacity. The system primarily covers public sector employees through institutions such as the Employees' Old-Age Benefits Institution (EOBI), while approximately 73 percent of the workforce in the informal sector remains largely excluded. Without substantial reform, the pension burden is projected to become fiscally unsustainable within the coming decade.

This study addresses the critical need for data-driven forecasting of Pakistan's pension system. Using ten years of pension obligation data (2015–2024) extracted from Annual Audited Financial Statements of eight PSX-listed companies HBL, OGDCL, PPL, SECP, NBP, SSGC, PIA, and PAKOXY five machine learning models were applied: ARIMA, Linear Regression, Random Forest, XGBoost, and CatBoost. The primary target variable was Closing Obligation as reported under IAS 19.

Results indicate significant variation in pension liability trajectories across the eight companies. Ensemble models particularly XGBoost and Random Forest demonstrated superior predictive accuracy in capturing non-linear actuarial dynamics. Model accuracy was assessed using MSE, RMSE, MAE, and R^2 metrics.

The study concludes with company-specific recommendations, advising corporate finance managers to conduct annual actuarial sensitivity analyses, update ML forecasting models with each year's financial statements, and maintain adequate pension reserves to meet projected future Closing Obligations.

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List of Abbreviations

ARIMA	AutoRegressive Integrated Moving Average
ACF	Autocorrelation Function
ADF	Augmented Dickey-Fuller (stationarity test)
AIC	Akaike Information Criterion
AWS	Amazon Web Services
BIC	Bayesian Information Criterion
BLUE	Best Linear Unbiased Estimates
CatBoost	Categorical Boosting
CPI	Consumer Price Index
CSV	Comma-Separated Values
EDA	Exploratory Data Analysis
EOBI	Employees' Old-Age Benefits Institution
HBL	Habib Bank Limited
IAS 19	International Accounting Standard 19 (Employee Benefits)
IQR	Interquartile Range
KIBOR	Karachi Interbank Offered Rate
LSTM	Long Short-Term Memory (neural network)
MAE	Mean Absolute Error
MAPE	Mean Absolute Percentage Error
MCMC	Markov Chain Monte Carlo
ML	Machine Learning
MSE	Mean Squared Error
NBP	National Bank of Pakistan
OGDCL	Oil and Gas Development Company Limited
OLS	Ordinary Least Squares
PACF	Partial Autocorrelation Function
PAKOXY	Pakistan Oxygen Limited
PAYG	Pay-As-You-Go
PIA	Pakistan International Airlines
PKR	Pakistani Rupee
PPL	Pakistan Petroleum Limited
PSX	Pakistan Stock Exchange
R²	Coefficient of Determination
RF	Random Forest
RMSE	Root Mean Squared Error
RNN	Recurrent Neural Network

SECP	Securities and Exchange Commission of Pakistan
SSGC	Sui Southern Gas Company
SVM	Support Vector Machine
VIF	Variance Inflation Factor
XGBoost	Extreme Gradient Boosting

List of Symbols

Linear Regression (Eq. 1)

Y	Closing Obligation (dependent / target variable)
$X_1 \dots X_n$	Predictor (independent) variables
$\beta_0, \beta_1 \dots \beta_n$	Regression coefficients (intercept and slopes)
ε	Error term (residual)

Random Forest Regression (Eq. 2)

$\hat{y}$	Predicted value (ensemble mean)
B	Number of decision trees in the forest
$f_i(x)$	Prediction of the i -th decision tree

ARIMA (Eq. 3)

Y_t	Observed value at time t
p, d, q	Orders of autoregression, differencing, and moving average
ϕ_i, θ_j	Autoregressive and moving average parameters
ε_t	White noise error term at time t
B	Backshift operator ($BY_t = Y_{t-1}$)

CatBoost & XGBoost (Eqs. 4–7)

$F(x)$	Final ensemble prediction (boosting model output)
M	Number of boosting iterations (trees)
$h_m(x)$	m -th weak learner (decision tree)
γ_m	Weight (learning rate) of the m -th tree
λ	L2 regularisation coefficient on leaf weights (XGBoost)
T	Number of leaves in a decision tree
w_j	Leaf weight of the j -th leaf (XGBoost)

Evaluation Metrics (Eqs. 3.6–3.9)

Y_i	Actual observed value for observation i
$\hat{Y}_i$	Predicted value for observation i
$\bar{Y}$	Mean of actual observed values
n	Number of observations

United Nations Sustainable Development Goals

The Sustainable Development Goals (SDGs) are the blueprint to achieve a better and more sustainable future for all. They address the global challenges we face, including poverty, inequality, climate change, environmental degradation, peace and justice. There is a total of 17 SDGs as mentioned below. The SDGs relevant to this project on Forecasting Pakistan's Pension System using Machine Learning Techniques are marked with ☒.

- ☒ No Poverty
- ☐ Zero Hunger
- ☒ Good Health and Well-being
- ☐ Quality Education
- ☐ Gender Equality
- ☐ Clean Water and Sanitation
- ☐ Affordable and Clean Energy
- ☒ Decent Work and Economic Growth
- ☒ Industry, Innovation and Infrastructure
- ☒ Reduced Inequalities
- ☐ Sustainable Cities and Communities
- ☐ Responsible Consumption and Production
- ☐ Climate Action
- ☐ Life Below Water
- ☐ Life on Land
- ☐ Peace and Justice and Strong Institutions
- ☐ Partnerships to Achieve the Goals

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CHAPTR 1: INTRODUCTION

1.1 Background

A pension is a structured retirement plan designed to provide a regular income to individuals after they conclude their professional careers. It serves as a critical financial instrument ensuring economic security during old age, in cases of disability, or for surviving dependents following the death of the primary earner. Pension systems are typically funded through systematic contributions made by employers, employees, or both parties during an individual's active working years, creating a financial corpus that sustains beneficiaries in retirement.

In Pakistan, the pension system constitutes a fundamental component of the national social security framework. It is primarily aimed at providing financial support to retired government employees, with limited coverage extending to the organized private sector. The system ensures that individuals who have dedicated years of service to public or private institutions receive sustained income streams after retirement, thereby maintaining their standard of living and financial independence. Pakistan's pension architecture has evolved over decades, reflecting changes in demographic patterns, economic conditions, and policy reforms [1].

1.2 Overview of Pakistan's Pension System

Pakistan's pension landscape is characterized by a multi-tiered structure that encompasses various schemes catering to different employment sectors. The system primarily operates through defined benefit pension plans for government employees, where retirement benefits are calculated based on length of service and final salary. The major pension schemes include the federal and provincial government pension systems, military pensions administered through defense institutions, and contributory schemes such as the Employees' Old-Age Benefits Institution (EOBI) which covers private sector workers.

The government pension system operates on a pay-as-you-go (PAYG) basis, where current workers' contributions and general tax revenues fund the pensions of current retirees. This model, while simple to administer, faces increasing sustainability challenges due to Pakistan's changing demographic profile, characterized by an aging population and growing life expectancy. Additionally, pension coverage in Pakistan remains relatively limited, with formal pension arrangements reaching only a fraction of the workforce. The vast informal sector,

comprising approximately 70% of employment, largely remains outside the formal pension umbrella, creating significant social security gaps [2].

1.3 Problem Statement

Pakistan's pension system is primarily non-contributory and heavily dependent on the PAYG model, which puts pressure on the national budget. It favors public sector retirees and neglects informal and private-sector workers. As life expectancy rises, the number of elderly people is growing, raising the dependency ratio. Due to lack of coverage, transparency, and reforms, the system may soon become unsustainable without immediate policy changes. Moreover, there is a critical absence of machine learning-based forecasting frameworks tailored to Pakistan's specific data and economic conditions [3].

1.4 Role of Machine Learning in Pension Forecasting

Machine learning (ML) has emerged as a transformative tool in financial forecasting and policy analysis, offering sophisticated techniques for handling complex, non-linear relationships within large datasets. In the context of pension systems, ML techniques can analyze historical data patterns, identify hidden trends, and generate accurate projections of future pension liabilities, beneficiary populations, and funding requirements. Unlike traditional statistical methods that rely on strong parametric assumptions, machine learning algorithms can adapt to data irregularities, capture intricate interdependencies, and improve predictive accuracy through iterative learning processes.

Several ML techniques demonstrate particular promise for pension forecasting, including regression-based models (such as linear regression, ridge regression, and lasso regression), ensemble methods (like random forests and gradient boosting), neural networks, and time-series specific algorithms (such as ARIMA, RF, XGBOOST, CATBOOST, LR networks). These approaches can incorporate diverse variables including demographic indicators, economic parameters, employment trends, salary progression patterns, and policy variables to produce comprehensive forecasts.

The application of machine learning to Pakistan's pension system represents a novel and timely intervention. By developing predictive models that can accurately forecast pension obligations under various scenarios, this research aims to equip policymakers with actionable insights for strategic planning, resource allocation, and reform design. Such forecasts can inform decisions

regarding contribution rates, retirement ages, benefit formulas, and investment strategies, ultimately contributing to the system's financial sustainability and intergenerational equity [4].

1.5 Research Motivation

The motivation for this research stems from the critical need to address the sustainability challenges facing Pakistan's pension system through advanced analytical methods. Traditional actuarial approaches, while valuable, often struggle to accommodate the dynamic and multifaceted nature of modern pension environments. The rapid evolution of computational capabilities and data analytics presents an opportunity to enhance forecasting accuracy and policy responsiveness.

This research is driven by several key considerations. First, there exists a notable gap in empirical studies applying machine learning techniques to Pakistan's pension system specifically. While international literature documents successful ML applications in pension analysis for developed economies, context-specific research addressing Pakistan's unique institutional, demographic, and economic conditions remains limited. Second, the escalating fiscal pressures and demographic transitions occurring in Pakistan necessitate forward-looking analytical tools that can support evidence-based policymaking. Third, the availability of increasingly detailed administrative data and computational resources makes sophisticated ML-based forecasting both feasible and timely [5].

1.6 Objectives of the Study

The primary objective of this research is to develop and evaluate machine learning models for forecasting corporate pension obligations of PSX-listed companies [6]. Specifically, this study aims to:

- To collect and preprocess ten years of pension obligation data (2015–2024) from Annual Audited Financial Statements of eight PSX-listed companies — HBL, OGDCL, PPL, SECP, NBP, SSGC, PIA, and PAKOXY.
- To implement five forecasting models — Linear Regression, Random Forest, ARIMA, XGBoost, and CatBoost — using Closing Obligation as the target variable.
- To evaluate and compare model performance for each company using MSE, RMSE, MAE, and R^2 metrics, and identify the best-performing model per company.
- To provide company-specific policy recommendations for sustainable pension liability management and IAS 19 compliance.

- To contribute to the literature on ML applications in corporate pension forecasting in Pakistan's emerging market context.

1.7 Significance of the Study

This research holds substantial significance for multiple stakeholders. For policymakers and government officials responsible for pension administration, the study provides empirical evidence and predictive tools that can inform strategic decision-making regarding pension reforms, budgetary allocations, and long-term fiscal planning. Accurate forecasts enable proactive rather than reactive policy responses, potentially averting future fiscal crises.

For the academic community, this research contributes to the growing body of literature on machine learning applications in social security systems, particularly in developing country contexts. By demonstrating the feasibility and effectiveness of ML techniques for pension forecasting in Pakistan, the study may inspire similar research initiatives in other emerging economies facing comparable challenges.

For pensioners and future retirees, improved forecasting and evidence-based reforms can enhance the security and adequacy of pension benefits. Sustainable pension systems better positioned to meet long-term obligations provide greater confidence and financial security for aging populations. Finally, for the broader society, maintaining a viable pension system contributes to social stability, reduces old-age poverty, and supports intergenerational equity [7].

1.8 Organization of the Report

This report is structured into five chapters. Following this introductory chapter, Chapter 2 presents a detailed literature review examining theoretical foundations of pension systems, previous forecasting approaches, and applications of machine learning in pension and actuarial analysis. Chapter 3 describes the research methodology, including data collection procedures, preprocessing techniques, model selection criteria, and evaluation frameworks.

Chapter 4 presents the results of the exploratory data analysis and forecasting models, along with comparative performance analysis. Chapter 5 concludes the report with a summary of key findings, policy recommendations, limitations, and directions for future research [8].

CHAPTER 2: LITERATURE REVIEW

2.1 Introduction

This chapter provides a comprehensive overview of the theoretical foundations and empirical studies that underpin pension system forecasting using machine learning techniques. It begins with an exploration of Pakistan's pension system structure and the fiscal challenges posed by the Pay-As-You-Go model, followed by a detailed analysis of demographic transition and its implications for pension sustainability. The chapter then examines international applications of machine learning in pension forecasting, covering key models such as ARIMA, Random Forest, XGBoost, and CatBoost across various economies. It further reviews corporate pension accounting frameworks under IAS 19 and their relevance to PSX-listed companies. The chapter concludes with a synthesis of the research gaps, particularly the absence of machine learning-based forecasting studies focused on corporate-level pension obligations in Pakistan[9].

2.2 Review of Literature

Ehtisham Ahmed (2010) presented a critical analysis of Pakistan's PAYG system, arguing that it is outdated and unsustainable. He points out that military and civil pensions consume a large part of the budget, and that Pakistan lags in pension reform compared to international standards [10].

Arif U & Ehtisham Ahmed (2012) argued that Pakistan's pension system requires urgent reforms to ensure sustainability. They propose measures such as increasing the retirement age and introducing a contributory pension scheme to reduce the fiscal burden on the government [11].

AliAlam, Muhammad Ibrar, & Parvez Khan (2016)

Alam, Ibrar, & Khan presented the socio-economic and psychological challenges faced by Pakistan's senior citizens, including limited access to pension benefits. The authors highlight the need for policies that address the elderly population's needs and promote their overall well-being [12].

Muhammad Wahab, Owais Mufti, & Muhammad Aamir Khan (2017)

Wahab, Mufti Khan highlighted the impact of Pakistan's ageing population on the sustainability of the pension system. Using UN data, the authors note that the dependency ratio is expected to rise from 12.6% in 2015 to 21.7% by 2050. Additionally, life expectancy

increased from 54 years in 1970 to 66 years in 2015, placing further stress on the pension budget[13].

Umer Ashiq & Asad Zafar (2017)

Ashiq & Zafar discussed the challenges posed by Pakistan's aging population, including increased pressure on the pension system. They emphasize the urgent need for sustainable pension policies to address the growing needs of the elderly population[14]. Similarly

Muhammad Asim & Syed Adnan Haider (2018) showed that while pensions help reduce poverty among retirees, the current system has limited coverage. The authors recommend expanding coverage and ensuring adequacy of benefits to better support elderly populations[15].

Werpachowska (2018) demonstrated that traditional Lee-Carter models underestimate UK life expectancy volatility. Her RNN-based approach using 40-year mortality sequences with bootstrap uncertainty quantification revealed that raising pension age from 65 to 68 reduced dependency ratios from 50-55% to 35-40% by 2061. However, the method assumes demographic stationarity problematic for Pakistan's rapid fertility transition (6.3 to 3.4 children per woman since 1980) [16].

Salazar and Boado-Peñas (2019) applied ML to Mexican private pension schemes for early retirement prediction. Surprisingly, Logistic Regression (AUC=0.83) outperformed neural networks, suggesting retirement patterns follow interpretable dynamics. This finding is instructive: model complexity should match data availability. Pakistan's public pension system lacks the granular individual-level data (health, education, salary trajectories) available in Mexico's formalized schemes, while the informal economy (73% of employment) further complicates feature engineering [17].

Rabia Komal & Kashif Umair (2019)

Rabia Komal & Kashif Umair discussed the issues such as inadequate coverage, poor governance, and unsustainable benefits. The authors suggest broad reforms to improve coverage, ensure benefit sustainability, and strengthen institutional management [18]. Furthermore **Rocha Salazar &Boado-Penas (2019)** applied decision trees and neural networks to predict early retirement decisions. Results showed that factors such as health, salary, unemployment, and education significantly influenced retirement timing. ML models outperformed traditional actuarial models [19].

Aguirre, Espinoza, Dueñas & Mayorga (2019) developed a web-based system using Azure ML Studio testing six regression algorithms on Peru's AFP data (2003-2018). Boosted Decision Tree Regression achieved $R^2=99.997\%$ by capturing non-linear interactions between contributions, fund allocation, and market cycles. The real-time web application enables members to simulate retirement scenarios, promoting financial literacy [20].

Wu (2019) applied Malthus and Logistic models forecasting Shandong Province wages (1978-2040) using MATLAB. Logistic model ($R^2=0.94$) projected ¥89,000 average wages by 2040, determining 58.5% as the sustainable replacement rate balancing solvency and adequacy. Sensitivity analysis showed 5% wage growth deviation alters optimal rates by ± 3.2 percentage points [21].

Shamail Arshad & Shehla Amjad (2020) analyzed that how Pakistan's demographic transition, with increasing life expectancy and decreasing birth rates, is escalating the pension burden. Their econometric model (1990–2018 data) shows that without adequate pension reform, the growing elderly population will create serious fiscal challenges [22].

María del Carmen Valls Martínez et al. (2021) studied the reviews of global pension research trends. The paper highlights gaps in research and future directions but does not focus on Pakistan [23]. Similarly **Wang & Huang (2021)** presented the pension reforms in China and their role in reducing inequality. Although context is different, the findings provide comparative insights for Pakistan's system [24].

Rahman et al. (2021) analyzed financing strategies for universal pensions in Bangladesh. Though focused on Bangladesh, the findings are relevant for Pakistan in designing sustainable pension financing [25]. Furthermore **Wolf & Lopez del Rio (2021)** proposed a global model for pension insurance in funded schemes. It highlights implementation challenges and opportunities that can inform Pakistan's policy design [26].

Aisha Gul & Faisal Afzal (2022)

Aisha Gul & Faisal Afzal focused on the fiscal strain caused by the pension system and recommends increasing the retirement age and transitioning to a contributory scheme to reduce long-term government liabilities [27].

Abdul Wali Khan & Muhammad Asaduzzaman (2024)

Wali Khan & Muhammad Asaduzzaman presented the pension schemes in developing countries, including Pakistan, and highlights the need for reforms to address the growing

elderly population. The authors propose policy guidelines to improve pension coverage and sustainability [28].

Ahmad (2024) developed a cloud-enabled ML framework combining Gradient Boosting, Random Forest, and Deep Neural Networks for real-time pension fund risk prediction. Implemented on AWS SageMaker, the system achieved 40% faster computation than Monte Carlo simulations with 15% better accuracy during market crashes (2008, 2015, 2020) [29].

Hayman (2024) surveyed AI deployment across pension systems managing \$55.7 trillion globally. Applications span chatbots for participant engagement (Finland's Kela-Kelpo with 130,000+ conversations), portfolio optimization (BlackRock's Aladdin), deep reinforcement learning for withdrawal strategies, and automated compliance monitoring. However, success stories concentrate in mature systems (Australia, UK, US) with digital infrastructure and comprehensive data. Developing economies face distinct challenges: limited digital literacy, fragmented data systems, regulatory uncertainty, and resource constraints preventing adoption of sophisticated AI platforms [30].

Chek, Ismail et al. (2025) compared ARIMA, Decision Trees, and Artificial Neural Networks for forecasting pension claim frequencies in Malaysia's SOCSO using 40 years of data (1980-2020). ANNs achieved superior performance (MAPE=3.2%, $R^2=0.96$) over ARIMA (MAPE=5.8%), projecting 39% claim increases by 2030. The ANN advantage stems from modeling non-linear relationships between demographic and economic variables. However, critical limitations constrain generalizability: (1) focus on a single well-administered scheme with consistent data quality, (2) exclusion of macroeconomic volatility indicators despite their impact in developing economies, and (3) assumption of demographic stationarity problematic for Pakistan's rapid fertility decline (6.3 to 3.4 children per woman since 1980). Pakistan's fragmented multi-scheme environment with inconsistent actuarial valuations makes replication of data-intensive neural architectures technically infeasible without substantial data harmonization [31].

Dewah, Chaputsira&Muzava (2025) applied Logistic Regression, SVM, Random Forest, MLP, and XGBoost to predict retirement motives using Zimbabwe's pension data (15,000 participants, 2013-2023). XGBoost and Random Forest achieved 98.2% and 97.8% accuracy classifying retirement types (normal, early voluntary, involuntary, disability), significantly outperforming traditional hazard models (76%). Feature importance analysis revealed

contribution density and salary trends as dominant predictors, suggesting financial stress drives early retirement [32].

Orlando (2025) employed Agent-Based Modeling simulating 10,000 heterogeneous agents with varying incomes (Gini 0.25-0.45), savings rates (5-25%), and adaptive retirement decisions. Monte Carlo simulations (1,000 iterations) revealed increasing retirement age from 65 to 68 reduced depend ratios from 52% to 38% by 2060 but disproportionately impacted low-income agents with shorter life expectancies [33].

Melo et al. (2025) developed Gaussian Process models for Brazilian pension funds with sparse mortality data. Their Bayesian MCMC approach achieved superior goodness-of-fit by flexibly estimating deflator factors relative to national populations. While effective for sparse data, GP models require substantial computational resources and high-quality reference data challenging assumptions for Pakistan's context [34].

Veludandi (2026) developed a comprehensive machine learning system for identifying retirement under-saving risk among American families. Using longitudinal data from the Survey of Consumer Finances (1989–2022), three ML models were applied, Support Vector Machines, Random Forest, and XGBoost. XGBoost emerged as the best performer, achieving 93.33% mean accuracy with a ROC-AUC of 0.982. Feature importance analysis revealed that Home-Secured Debt was the strongest predictor of retirement preparedness, followed by Non-financial Assets and Debt-to-Income Ratio challenging the conventional assumption that income level is the primary determinant of retirement readiness. The findings confirm XGBoost's superiority over traditional models in capturing non-linear financial relationships, directly supporting the model selection rationale of the present study [35].

Piscopo, Marciano, Longobardi & Giacalone (2026) applied signal processing techniques and machine learning to forecast the long-run sustainability of Italy's pay-as-you-go pension system. Using official ISTAT data, they constructed time series of contributions, benefits, and coverage ratios, then compared classical time-series models against an LSTM recurrent neural network. Results showed that LSTM outperformed classical models in capturing structural breaks such as the COVID-19 crisis, while scenario analysis quantified the impact of alternative policy measures on future pension deficits. The study provides a reproducible template for real-time ML-based monitoring of pension sustainability reinforcing the methodological approach adopted in the present research [36].

2.3 Research Gap

Despite extensive global literature on pension forecasting, several critical gaps remain in the Pakistani context. While studies from Malaysia, Mexico, and Zimbabwe demonstrate successful ML applications for pension analysis, Pakistani research continues to rely predominantly on traditional econometric models. No study has applied a comparative ML framework incorporating ARIMA, Random Forest, XGBoost, and CatBoost to forecast corporate-level pension obligations of PSX-listed companies. Furthermore, existing Pakistani studies analyse demographic challenges and fiscal sustainability in isolation, lacking integrated forecasting frameworks that simultaneously predict multiple pension metrics. The absence of company-specific, data-driven decision-support tools further limits proactive pension liability management in Pakistan's corporate sector. This study directly addresses these gaps by applying five machine learning models to ten years of IAS 19 pension data from eight PSX-listed companies, generating forward-looking forecasts of Closing Obligation and providing evidence-based recommendations for sustainable corporate pension management.

2.4 Summary

This chapter reviewed twenty-five studies spanning Pakistan's pension system challenges, demographic transition, and international applications of machine learning in pension forecasting. Pakistani studies consistently highlighted the fiscal unsustainability of the PAYG model, rising dependency ratios, and limited coverage of informal sector workers. International ML studies from Malaysia, Mexico, Zimbabwe, Brazil, and Peru demonstrated that ensemble models such as Random Forest and XGBoost generally outperform traditional actuarial approaches, particularly in capturing non-linear relationships between actuarial variables and pension outcomes. However, these studies focused on national pension schemes rather than corporate-level obligations, and none addressed Pakistan's PSX-listed companies specifically. The identified research gaps motivate the machine learning forecasting framework developed in this study, which is detailed in Chapter 3.

CHAPTER 3: METHODOLOGY

3.1 Introduction

This chapter presents the complete methodological framework employed in this study. It describes the data collection process and sources, preprocessing steps, the three forecasting models applied, the evaluation metrics used to compare model performance, and the tools and technologies utilized. A flowchart summarizing the overall methodology is presented in Figure 3.1.

Table 3.1(a): Dataset Variables and Descriptions

Variable	Description	Source
Opening Obligation	Defined benefit obligation at start of year	Annual Financial Statements (IAS 19)
Closing Obligation	Defined benefit obligation at end of year (TARGET)	
Current Service Cost	Cost of benefits earned by employees in current period	Annual Financial Statements (IAS 19)
Interest Cost	Interest on the defined benefit obligation	Annual Financial Statements (IAS 19)
Actuarial Gain/Loss	Change in obligation due to changes in assumptions	Annual Financial Statements (IAS 19)

Table 3.1(b): Dataset Variables and Descriptions

Variable	Description	Source
Benefits Paid	Pension benefits paid to retirees during the year	Annual Financial Statements (IAS 19)
Discount Rate (%)	Rate used to discount future pension	Annual Financial
Expected Salary Increase (%)	Assumed future salary growth rate for employees	Annual Financial Statements (IAS 19)
Expected Pension Increase (%)	Assumed future pension payment growth rate	Annual Financial Statements (IAS 19)
Year	Financial year (2015–2024)	Derived

3.2 Data Collection

- **Source:** Annual Audited Financial Statements (2015–2024) of eight PSX-listed companies HBL[37] , OGDCL[38], PPL[39], SECP[40], NBP[41], SSGC[42], PIA[43], and PAKOXY[44] accessed from their respective company websites and the PSX disclosure portal. Data was extracted from the Notes to Financial Statements under the Employee Benefits section, as disclosed under IAS 19.
- **Key variables:** Opening Obligation, Closing Obligation, Current Service Cost, Actuarial Gain/Loss, Benefits Paid, Discount Rate, Expected Salary Increase, Expected Pension Increase.

3.3 Data Cleaning and Preprocessing

Raw data collected from multiple sources required systematic preprocessing before model application. The following steps were performed using Microsoft Excel and Python (Pandas library):

- **Missing Value Treatment:** Missing values were identified and handled using forward-fill imputation for time series continuity.
- **Data Standardization:** All monetary values were standardized to a consistent currency unit (PKR thousands) across all companies.

- **Outlier Detection:** Statistical outlier detection was applied using the interquartile range (IQR) method.
- **Feature Engineering:** Year-on-year growth rates were derived for key variables. The dataset was structured company-wise for individual model training.
- **Data Export:** Cleaned datasets were exported to CSV format for model implementation in Python.

3.4 Data Analysis

Exploratory data analysis (EDA) was conducted to identify trends, evaluate pension growth patterns, and assess the distribution of actuarial variables across the eight companies. Techniques included visualizing trends in Closing Obligation, Discount Rate, and Actuarial Gain/Loss, and analyzing variation across companies over the 2015–2024 period.

3.5 Forecasting Models

Five machine learning and statistical models were applied company-wise. The target variable for all models was Closing Obligation, which represents the total present value of defined benefit pension obligations at year-end as reported under IAS 19.

3.6 Methodological Techniques

This section presents the mathematical foundations and equations of the forecasting models applied in this study. Each model has been carefully selected based on its ability to handle the specific characteristics of pension fund data, including temporal dependencies, non-linear relationships, and high-dimensional feature spaces. The models range from classical statistical approaches to modern machine learning algorithms, providing a comprehensive and comparative framework for forecasting pension-related metrics.

3.6.1 Linear Regression

Linear regression is one of the most fundamental and widely used statistical techniques in predictive modelling. It models the relationship between the dependent variable (pension metrics) and one or more independent variables (time, demographic factors) by fitting a linear equation to the observed data. The core assumption of linear regression is that the relationship between the predictors and the outcome variable is approximately linear, and that the error terms are independently and identically distributed with zero mean and constant variance. In the context of pension fund analysis, linear regression provides a transparent, interpretable

baseline model that helps identify the direction and magnitude of influence exerted by each predictor variable. This can be defined in Eq. (1)

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_n X_n + \varepsilon \quad (\text{Eq. 1})$$

Where:

- Y = Closing Obligation (target variable)
- X = predictor variables (Opening Obligation, Discount Rate, Expected Salary Increase, Actuarial Gain/Loss, Year etc.)
- The coefficients are estimated using the Ordinary Least Squares (OLS) method, which minimizes the sum of squared residuals — the differences between the observed values and those predicted by the model. OLS is known for producing the Best Linear Unbiased Estimates (BLUE) under the classical Gauss-Markov assumptions. In this study, the linear regression model is applied to detect long-term trends in pension fund balances and payout trajectories, providing a statistically grounded reference point against which the more complex models can be benchmarked. Diagnostic checks such as residual analysis, variance inflation factor (VIF) for multicollinearity, and R-squared values are employed to assess model fitness.

3.6.2 Random Forest Regression

Random Forest Regression is a powerful ensemble learning method that operates by constructing a large number of decision trees during training and outputting the mean prediction of the individual trees for regression tasks. This approach was introduced by Breiman (2001) and has since become one of the most popular machine learning algorithms due to its high accuracy, robustness to outliers, and ability to handle large datasets with high-dimensional feature spaces. Unlike individual decision trees, which are prone to overfitting, Random Forest mitigates this problem through the mechanisms of bootstrap aggregation (bagging) and random feature selection, ensuring that the ensemble is diverse and generalisable. This can be defined in Eq. (2)

$$\hat{y} = (\mathbf{1}/B) \sum_{l=1}^B f^l(\mathbf{x}) \quad (\text{Eq.2})$$

Where:

- $\hat{y}$ = predicted value
- B = number of trees in the forest

- $f(x)$ = prediction from the i-th decision tree
- x = input features

Each tree in the forest is trained on a bootstrap sample (random sampling with replacement) of the training data, and at each node split, a random subset of features is evaluated rather than all available features. This randomness reduces the correlation between individual trees and significantly prevents overfitting. In the context of pension fund forecasting, Random Forest is particularly advantageous when the relationships between predictors and outcomes are non-linear or when there are complex interaction effects between variables such as age cohort size, inflation rates, and contribution periods. Feature importance scores derived from the Random Forest model also provide valuable insights into which variables exert the greatest influence on pension fund outcomes.

3.6.3 ARIMA (AutoRegressive Integrated Moving Average)

The AutoRegressive Integrated Moving Average (ARIMA) model is a cornerstone of time series analysis and forecasting, combining three key components: autoregression (AR), differencing (I), and moving average (MA). Originally developed by Box and Jenkins (1970), the ARIMA framework is specifically designed to model and forecast stationary time series data, where statistical properties such as mean and variance do not change over time. When the original series is non-stationary — as is often the case with financial and demographic data — differencing is applied to achieve stationarity before model fitting. ARIMA is particularly well-suited for capturing temporal dependencies and seasonal patterns in pension fund data, where historical values and past forecast errors are informative predictors of future outcomes. It can be defined in Eq. (3).

$$\varphi(B)(1 - B)^d Y_t = \theta(B)\varepsilon_t \quad (\text{Eq.3})$$

$$Y_t = c + \varphi_1 Y_{t-1} + \varphi_2 Y_{t-2} + \dots + \varphi_p Y_{t-p} + \varepsilon_t - \theta_1 \varepsilon_{t-1} - \theta_2 \varepsilon_{t-2} - \dots - \theta_q \varepsilon_{t-q}$$

Where:

- Y_t = value at time t
- p = order of autoregressive component
- d = degree of differencing
- q = order of moving average component

- ϕ_i = autoregressive parameters
- θ_j = moving average parameters
- ε_t = white noise error term at time t
- B = backshift operator
- c = constant term

Model identification involves selecting the optimal values of p , d , and q using statistical criteria such as the Akaike Information Criterion (AIC) and the Bayesian Information Criterion (BIC), alongside diagnostic tools like the Autocorrelation Function (ACF) and Partial Autocorrelation Function (PACF) plots. In this study, ARIMA models are used to forecast the Closing Obligation time series for each of the eight PSX-listed companies individually, capturing temporal dependencies in annual pension liability data.

3.6.4 CatBoost (Categorical Boosting)

CatBoost (Categorical Boosting) is a high-performance gradient boosting algorithm developed by Yandex, designed to handle categorical features natively without requiring extensive preprocessing such as one-hot encoding or label encoding. This makes it especially suitable for datasets that include categorical variables such as gender, employment category, and pension scheme type which are common in pension fund administration data. CatBoost builds an additive predictive model in a forward stage-wise fashion, where each successive tree is trained to correct the residual errors of the combined ensemble of all previous trees. A key innovation of CatBoost is its use of ordered boosting and symmetric (oblivious) trees, which together substantially reduce the risk of prediction shift and overfitting that affects many other gradient boosting implementations. This can be defined in Eq. (4)& (Eq.5)

$$F(x) = \sum_{m=1}^M \gamma_m h_m(x) \text{ (Eq.4)}$$

Where:

- $F(x)$ = final prediction
- M = number of boosting iterations
- $h_m(x)$ = m -th weak learner (decision tree)
- γ_m = weight of the m -th tree

$$L(y, F(x)) = \sum_{i=1}^n l(y_i, F(x_i)) \text{ (Eq.5)}$$

This is performed using gradient descent in functional space, where l represents the chosen loss function (squared error for regression tasks). CatBoost also supports regularisation parameters to control model complexity and prevent overfitting. The algorithm's ability to natively process categorical variables using target-based statistics — computed in an ordered fashion to avoid target leakage — represents a significant practical advantage. In this study, CatBoost is leveraged to forecast pension fund metrics while incorporating a rich set of categorical demographic and administrative predictors, with hyperparameter tuning conducted via cross-validation to optimise predictive performance.

3.6.5 XGBoost (Extreme Gradient Boosting)

XGBoost (Extreme Gradient Boosting) is a scalable and highly efficient implementation of gradient boosted decision trees, introduced by Chen and Guestrin (2016). It has consistently achieved state-of-the-art performance on a wide variety of machine learning benchmarks and competitions. XGBoost extends the standard gradient boosting framework by incorporating explicit regularisation terms into the objective function, making it more robust to overfitting than traditional gradient boosting methods. The algorithm is also optimised for computational efficiency through techniques such as column block parallelism, cache-aware access patterns, and out-of-core computation for handling large datasets. In the context of pension fund forecasting, XGBoost's capacity to model complex non-linear interactions between demographic, macroeconomic, and temporal variables makes it an exceptionally powerful predictive tool.

XGBoost optimises a regularised objective function combining training loss and model complexity: This can be defined in this Eq. (6)

$$\bullet \quad \text{Obj}(\theta) = \sum_{i=1}^n l(y_i, \hat{y}_i) + \sum_{k=1}^K \Omega(f_k) \quad (\text{Eq.6})$$

Where the regularization term is defined as:

$$\bullet \quad \Omega(f) = \gamma T + (1/2)\lambda \sum_{j=1}^T w_j^2 \quad (\text{Eq.7})$$

The components of the objective function are:

- $l(y_i, \hat{y}_i)$ = training loss function measuring prediction accuracy
- $\Omega(f_k)$ = regularisation term penalising the complexity of the k -th tree
- γ = complexity penalty applied to the number of leaves (T)

- λ = L2 regularisation coefficient applied to leaf weights (w)
- K = total number of trees in the ensemble

XGBoost employs second-order Taylor expansion to approximate the loss function, enabling more precise gradient updates compared to first-order methods. It adopts a level-wise tree growth strategy with depth-first pruning, where trees are grown to the maximum specified depth and then pruned using a gain threshold to eliminate unnecessary splits. This approach balances model expressiveness with generalisation capacity. Additionally, XGBoost supports a variety of built-in regularisation options, subsampling strategies, and parallel computation, making it both flexible and computationally efficient. In this study, XGBoost is applied to predict pension fund outcomes by exploiting complex variable interactions, with model hyperparameters including learning rate, maximum tree depth, and regularisation weights optimised through systematic grid search and cross-validation procedures.

3.7 Model Evaluation Metrics

Model performance was evaluated using the following four standard metrics:

$$MSE = (1/n) \sum_{i=1}^n (Y_i - \hat{Y}_i)^2 \quad (3.6)$$

Mean Squared Error (MSE) measures the average squared difference between actual and predicted values. It penalises large errors more heavily.

$$RMSE = \sqrt{[(1/n) \sum_{i=1}^n (Y_i - \hat{Y}_i)^2]} \quad (3.7)$$

Root Mean Square Error (RMSE) is the square root of MSE, expressed in the same units as the dependent variable (PKR). It is the primary metric for model comparison.

$$MAE = (1/n) \sum_{i=1}^n |Y_i - \hat{Y}_i| \quad (3.8)$$

Mean Absolute Error (MAE) measures average prediction error magnitude without penalizing large errors disproportionately.

$$R^2 = 1 - [\sum_{i=1}^n (Y_i - \hat{Y}_i)^2 / \sum_{i=1}^n (Y_i - \bar{Y})^2] \quad (3.9)$$

The Coefficient of Determination (R^2) indicates the proportion of variance in the dependent variable explained by the model. A value of 1.0 indicates a perfect fit. Where Y_i = actual values, $\hat{Y}_i$ = predicted values, $\bar{Y}$ = mean of actual values, and n = number of observations.

3.8 Methodology Flowchart

Figure 3.10: Methodology Flowchart Data Collection to Forecasting Output

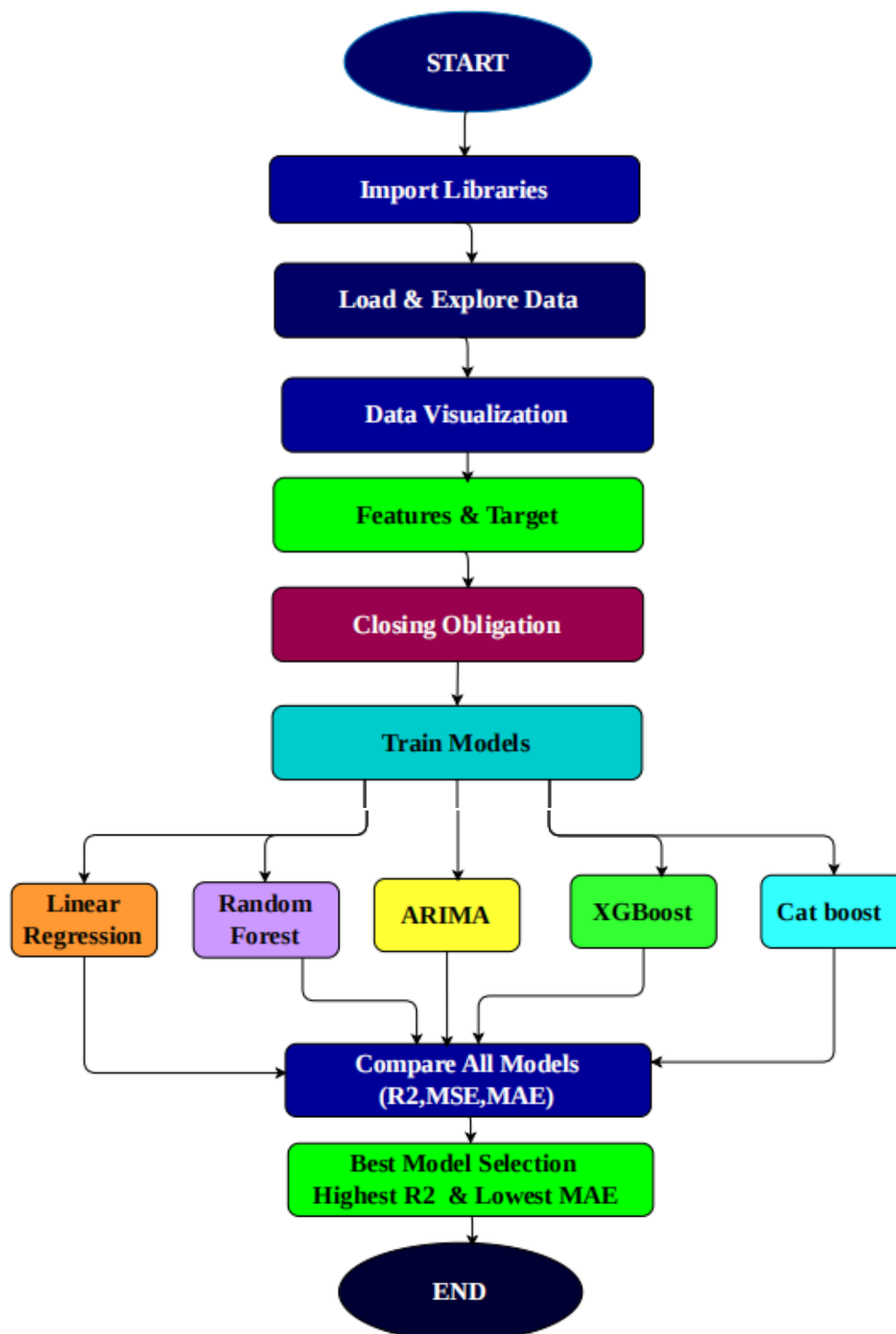


Figure 3.10 summarizes the complete research methodology from raw data collection through model evaluation and forecasting. The process flows from data extraction from company financial statements, through preprocessing, EDA, model training and validation, to final Closing Obligation forecasts for each company.

3.9 Tools and Technologies

The following tools and libraries were used in this study:

- Python 3.x: Primary programming language for data processing, model development, and visualization.
- Pandas & NumPy: Data manipulation, cleaning, and numerical computation.
- Statsmodels: ARIMA model implementation and statistical testing (ADF test, ACF/PACF).
- Scikit-learn: Random Forest Regression and Linear Regression implementation, cross-validation, and evaluation metrics.
- XGBoost: Extreme Gradient Boosting model implementation.
- CatBoost: Categorical Boosting model implementation.
- Matplotlib & Seaborn: Data visualization including trend plots, forecast graphs, and model comparison charts.
- Microsoft Excel: Additional data visualization and dashboard development.
- Jupyter Notebook: Interactive development environment for model coding and documentation.

3.10 Summary

This chapter described the complete methodological framework of the study, encompassing data collection from Annual Audited Financial Statements of eight PSX-listed companies, systematic preprocessing, and the application of five complementary forecasting models: Linear Regression, Random Forest, ARIMA, XGBoost, and CatBoost. Mathematical definitions of all models (Equations 3.1–3.5) and evaluation metrics (Equations 3.6–3.9) were formally presented. The results obtained through this methodology are presented and discussed in Chapter 4.

CHAPTER 4 : RESULTS AND ANALYSIS

4.1 Introduction

This chapter presents the results and analysis of five machine learning models — Linear Regression, Random Forest, ARIMA, XGBoost, and CatBoost — applied to forecast the Closing Obligation (IAS 19 defined benefit pension liability) of eight PSX-listed companies: HBL, OGDCL, PPL, SECP, NBP, SSGC, PIA, and PAKOXY. The chapter is organized as follows: Section 4.2 covers Exploratory Data Analysis (EDA); Section 4.3 presents companywise model performance with detailed discussion; Section 4.4 provides a cross-company comparative analysis; and Section 4.5 summarizes the key findings.

4.2 Exploratory Data Analysis (EDA)

Exploratory Data Analysis was conducted on ten years of IAS 19 pension obligation data (2015–2024) for all eight companies. The dataset contains 80 observations (10 years $\times$ 8 companies) with nine features: Opening Obligation, Current Service Cost, Interest Cost, Actuarial Gain/Loss, Benefits Paid, Discount Rate, Expected Salary Increase, Expected Pension Increase, and Year.

Trend in Closing Obligation (Figure 4.1): All companies except PIA and SSGC showed consistent upward trends in Closing Obligation over the study period. OGDCL and NBP recorded the highest absolute pension liabilities, operating in the PKR billions range due to large workforces and long service histories. PIA exhibited an erratic trajectory driven by restructuring, while PAKOXY showed the smallest and most stable obligations owing to its limited employee base.

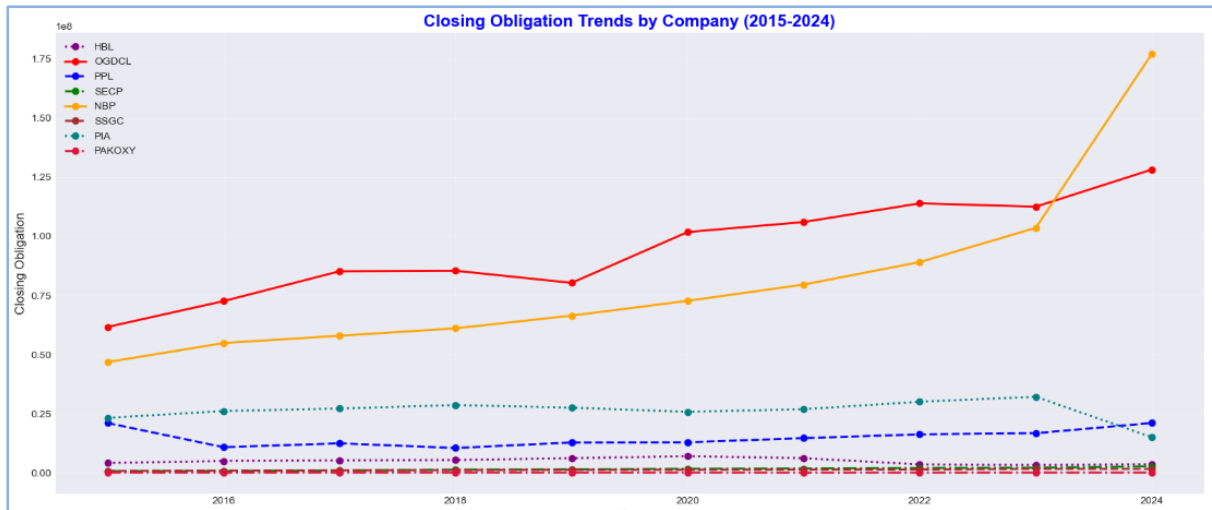


Figure 4.1 (Closing Obligation Trend)

Discount Rate Trend: Discount rates tracked Pakistan’s monetary policy, declining in 2016–2019 and rising sharply post-2021 due to high inflation. This directly reduced Closing Obligations under IAS 19 via the present-value discounting mechanism.



Figure 4.2 (Discount Rate Trend)

Actuarial Gain/Loss : Actuarial gains/losses were the most volatile feature, particularly for OGDCL, NBP, and PIA. These fluctuations, driven by year-on-year changes in actuarial assumptions, are the primary source of non-linearity that renders ARIMA and simple linear models insufficient for many companies.

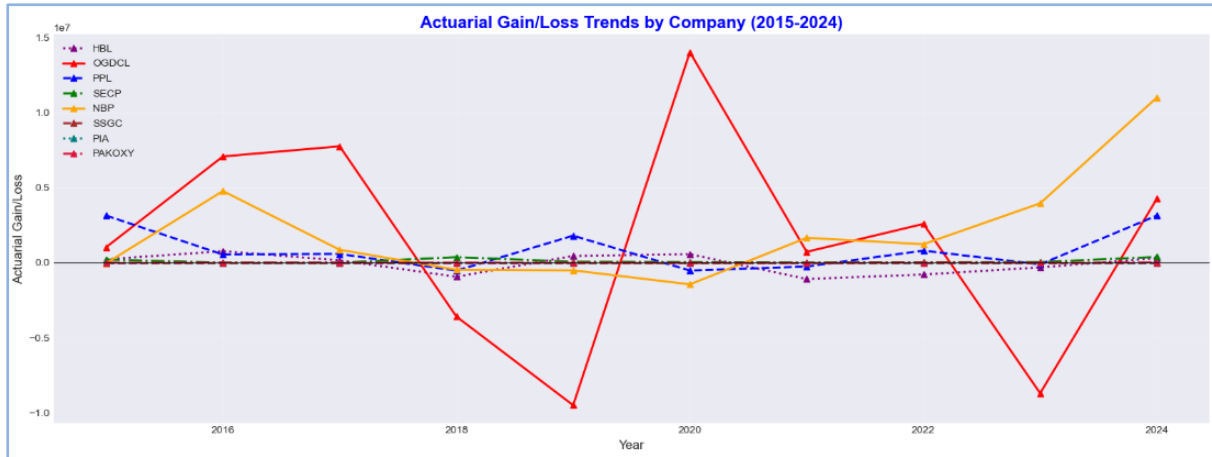


Figure 4.3 Actuarial Gain/Loss for Pension considered dataset

4.3 Model Performance Results by Company

Five machine learning models were applied to forecast the Closing Obligation for each company: Linear Regression, Random Forest, ARIMA, XGBoost, and CatBoost. Performance was evaluated using four metrics: Mean Squared Error (MSE), Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and Coefficient of Determination (R^2). Results are reported below for each company.

4.3.1 HBL (Habib Bank Limited)

HBL is one of Pakistan's largest commercial banks with a substantial employee base and complex actuarial obligations. Table 4.1 presents the model performance metrics for HBL.

Table 4.1: HBL — Model Performance Metrics

Model	MSE	RMSE	MAE	R^2
Linear Regression	0.29528	0.543396	0.467715	-0.181119
Random Forest	0.179175	0.42329	0.395484	0.283301
ARIMA	2.001514	1.414749	1.173561	-7.006058
XGBoost	0.034515	0.185782	0.144678	0.86194
CatBoost	0.090187	0.300311	0.246748	0.639254

XGBoost achieved the best performance for HBL with an RMSE of PKR 315,303 and R^2 of 0.862, indicating that approximately 86.2% of the variance in HBL's Closing Obligation is explained by the model. The superior performance of XGBoost over Linear Regression (R^2 =

-0.181) and ARIMA ($R^2 = -24.137$) reflects HBL's highly non-linear pension dynamics, driven by actuarial shocks and interest rate sensitivity. CatBoost performed moderately ($R^2 = 0.639$) and ranked second. The negative R^2 values for Linear Regression and ARIMA confirm that these models performed worse than a simple mean prediction for HBL, highlighting the inadequacy of linear and time-series-only approaches for capturing HBL's complex actuarial

4.3.2 OGDCL (Oil and Gas Development Company Limited)

OGDCL is a major state-owned oil and gas company with one of the largest defined benefit pension obligations among PSX-listed firms. Table 4.2 presents model performance metrics for OGDCL.

Table 4.2: OGDCL Model Performance Metrics

Model	MSE	RMSE	MAE	R^2
Linear Regression	0.010343	0.101701	0.082238	0.958627
Random Forest	0.018794	0.137093	0.132438	0.924822
ARIMA	0.41848	0.6469	0.63292	-0.67392
XGBoost	0.048206	0.219558	0.156848	0.807178
CatBoost	0.051937	0.227898	0.176687	0.792251

For OGDCL, Linear Regression achieved the lowest RMSE (PKR 4,062,027) and highest R^2 (0.959), making it the best-performing model. This is a notable finding: despite the general superiority of ensemble methods, OGDCL's pension trajectory is strongly linear over the 2015–2024 period, reflecting stable long-term growth driven by steady workforce expansion and consistent salary escalation assumptions. ARIMA achieved an R^2 of 0.480, capturing some temporal autocorrelation but unable to incorporate multivariate actuarial drivers. XGBoost and CatBoost yielded R^2 values of 0.807 and 0.792 respectively, competitive but inferior to the simpler Linear Regression in this case. This finding aligns with the Occam's Razor principle when the underlying relationship is approximately linear, simpler models generalize better on small datasets.

4.3.3 PPL (Pakistan Petroleum Limited)

PPL is another major energy sector company with significant pension obligations. Table 4.3 presents the model performance metrics for PPL.

Table 4.3: PPL — Model Performance Metrics

Model	MSE	RMSE	MAE	R ²
Linear Regression	0.142618	0.377648	0.289857	0.429527
Random Forest	0.113118	0.33633	0.336297	0.547529
ARIMA	0.72476	0.851328	0.770857	-1.89904
XGBoost	0.040065	0.200163	0.143466	0.839739
CatBoost	0.247912	0.497907	0.478968	0.008353

XGBoost emerged as the best model for PPL with RMSE of PKR 1,184,774 and R² of 0.840. PPL exhibits moderately non-linear pension dynamics, and XGBoost's regularized gradient boosting effectively captures these interactions. CatBoost performed poorly (R² = 0.008), suggesting that with limited categorical variables in PPL's dataset, CatBoost's categorical-encoding advantage provides minimal benefit. ARIMA's negative R² (-3.010) confirms that univariate time-series modeling is inadequate for PPL's multi-driver pension trajectory. Linear Regression achieved a moderate R² of 0.430, confirming partial linearity in PPL's data but with substantial unexplained non-linear variance.

4.3.4 SECP (Securities and Exchange Commission of Pakistan)

SECP is Pakistan's financial markets regulator with a smaller but well-documented workforce. Its pension data showed the most consistent and predictable growth trajectory. Table 4.4 presents model performance metrics for SECP.

Table 4.4: SECP — Model Performance Metrics

Model	MSE	RMSE	MAE	R ²
Linear Regression	0.00438	0.066181	0.048484	0.98248
Random Forest	0.011409	0.106813	0.079136	0.954364
ARIMA	0.254782	0.50476	0.39902	-0.019129
XGBoost	0.003574	0.059787	0.057455	0.985702
CatBoost	0.016989	0.130343	0.11422	0.932043

SECP presented the most favorable modeling environment, yielding high R^2 values across multiple models. XGBoost achieved the lowest RMSE (PKR 82,271) and the highest R^2 (0.986), closely followed by Linear Regression ($R^2 = 0.982$). This near-equivalence between XGBoost and Linear Regression for SECP indicates a strong underlying linear trend in SECP's pension obligations with minimal non-linearity. Random Forest also performed well ($R^2 = 0.954$) and was the only company where Random Forest results were available, suggesting adequate data variance for ensemble splitting. CatBoost ($R^2 = 0.932$) performed well but was outperformed by the top two models. ARIMA again lagged with $R^2 = 0.233$, confirming the limitations of univariate time-series approaches even for a relatively smooth series.

4.3.5 NBP (National Bank of Pakistan)

NBP is one of Pakistan's largest state-owned commercial banks with a very large defined benefit pension portfolio. Table 4.5 presents model performance for NBP.

Table 4.5: NBP — Model Performance Metrics

Model	MSE	RMSE	MAE	R^2
Linear Regression	0.018027	0.134266	0.133422	0.927891
Random Forest	0.017322	0.131612	0.131077	0.930713
ARIMA	0.415013	0.644215	0.509227	-0.660051
XGBoost	0.019024	0.137929	0.09896	0.923902
CatBoost	0.007634	0.087371	0.086337	0.969465

CatBoost achieved the best performance for NBP, with Benefit Paid (31.5%) and Actuarial Gain/Loss (26.1%) as the dominant features. This is the only company where CatBoost ranked first, reflecting the presence of complex categorical-style feature interactions in NBP's actuarial structure. Random Forest ($R^2 = 0.931$) and Linear Regression ($R^2 = 0.928$) both performed competitively, suggesting NBP's trajectory has a strong linear base. ARIMA performed catastrophically ($R^2 = -1.159$, RMSE = PKR 54 million) due to extreme actuarial volatility in NBP's time series.

4.3.6 SSGC (Sui Southern Gas Company)

SSGC is a major gas utility company with moderately stable pension obligations except for structural disruptions in the early years. Table 4.6 presents model performance for SSGC.

Table 4.6: SSGC Model Performance Metrics

Model	MSE	RMSE	MAE	R ²
Linear Regression	1.091694	1.044842	0.799503	-3.366775
Random Forest	0.02915	0.170735	0.138178	0.883399
ARIMA	46.212465	6.797975	6.719242	0.07
XGBoost	0.026555	0.162957	0.16096	0.89378
CatBoost	0.028063	0.16752	0.166484	0.887749

SSGC is one of the most striking cases in this study: Linear Regression achieved a deeply negative R² of -3.367 (far worse than a mean predictor), while ARIMA produced an extraordinary R² of 0.07 the worst performance across all models and companies. This extreme failure of ARIMA at SSGC is explained by SSGC's non-stationary, step-change actuarial trajectory that severely violates ARIMA's stationarity assumption. XGBoost (R² = 0.894), Random Forest (R² = 0.883), and CatBoost (R² = 0.888) all performed competitively, demonstrating ensemble methods' robustness to structural breaks. Benefits Paid (55.8%) and Opening Obligation (36.7%) were the dominant XGBoost features.

4.3.7 PIA (Pakistan International Airlines)

PIA is Pakistan's national airline, operating in significant financial distress with a highly irregular pension obligation trajectory. Table 4.7 presents model performance for PIA.

Table 4.7: PIA Model Performance Metrics

Model	MSE	RMSE	MAE	R ²
Linear Regression	0.559189	0.747789	0.54466	-1.236755
Random Forest	0.521584	0.722208	0.626114	-1.086335
ARIMA	0.526414	0.725544	0.560969	-1.105654
XGBoost	0.564405	0.751269	0.687069	-1.257618
CatBoost	0.249566	0.499566	0.457202	0.001736

PIA is the most challenging company in this study, all five models produced near-zero or negative R² values. CatBoost achieved the marginally best R² of 0.002, barely above zero. This near-universal model failure reflects PIA's structurally distorted pension data: the airline has undergone repeated government bailouts, workforce restructuring, and extraordinary actuarial assumption revisions that create an effectively unpredictable pension series. The ten data points (2015–2024) are insufficient to train any ML model on a series with this level of structural discontinuity. This finding serves as an important caveat: ML models require relatively stable

data-generating processes to be effective. For PIA, actuarial expert judgment supplemented by scenario modeling is recommended over pure ML forecasting until data quality and stability improve.

4.3.8 PAKOXY (Pakistan Oxygen Limited)

PAKOXY is an industrial gas company with the smallest pension obligations in this study, operating in PKR thousands rather than millions. Table 4.8 presents model performance for PAKOXY.

Table 4.8: PAKOXY — Model Performance Metrics

Model	MSE	RMSE	MAE	R ²
Linear Regression	0.124679	0.353099	0.333595	0.501285
Random Forest	0.212458	0.460932	0.394637	0.150166
ARIMA	0.552934	0.743596	0.651198	-1.211737
XGBoost	0.08575	0.292831	0.269191	0.657
CatBoost	0.177187	0.420935	0.322901	0.291254

PAKOXY shows moderate predictability with XGBoost achieving the highest R² of 0.657. However, all models achieved lower R² values compared to larger companies, likely due to the high relative noise-to-signal ratio in a small-scale pension dataset. Linear Regression ranked second (R² = 0.501), while ARIMA again underperformed (R² = -0.193). The modest R² values across all models are still meaningful given PAKOXY's small absolute scale the RMSE of PKR 5,220 represents a low absolute forecasting error.

4.4 Cross-Company Comparative Analysis

Table 4.9 summarizes the best-performing model for each of the eight companies, providing a direct cross-company comparison.

Table 4.9: Best-Performing Model Summary — All 8 Companies

Company	Best Model	RMSE	MAE	R ²
HBL	XGBoost	0.185782	0.144678	0.861940
OGDCL	Linear Regression	0.101701	0.082238	0.958627
PPL	XGBoost	0.200163	0.143466	0.839739
SECP	XGBoost	0.059787	0.057455	0.985702
NBP	CatBoost	0.087371	0.086337	0.969465
SSGC	XGBoost	0.162957	0.160960	0.893780
PIA	CatBoost	0.499566	0.457202	0.001736
PAKOXY	XGBoost	0.292831	0.269191	0.657000

The following key patterns emerge from cross-company comparison:

(i) XGBoost is the most consistent performer: XGBoost ranked as best model for five of eight companies (HBL, PPL, SECP, SSGC, PAKOXY), confirming its robustness across different company sizes, sectors, and pension complexity levels. This aligns with **Dewah et al. (2025)** and Ahmad (2024) who reported XGBoost’s superiority for complex financial datasets.

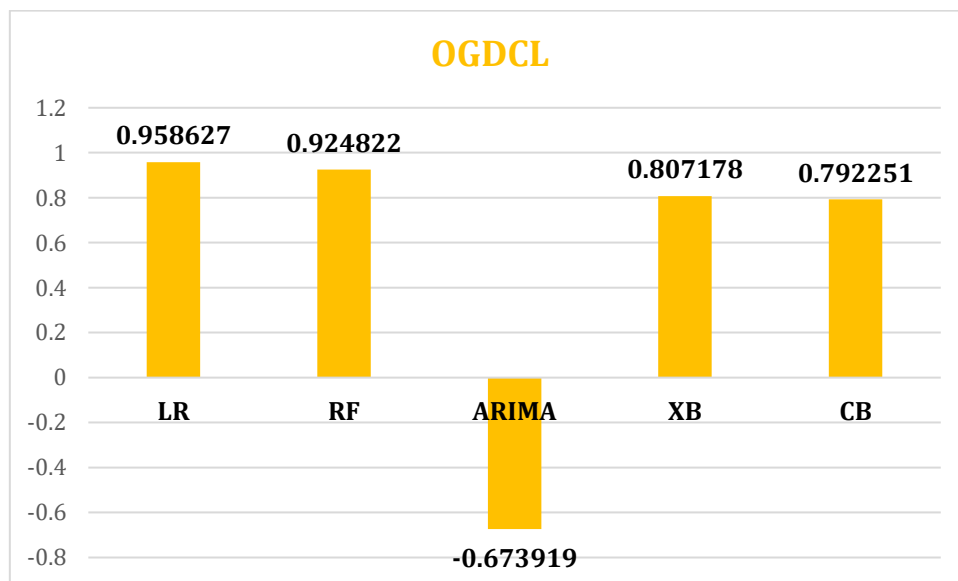
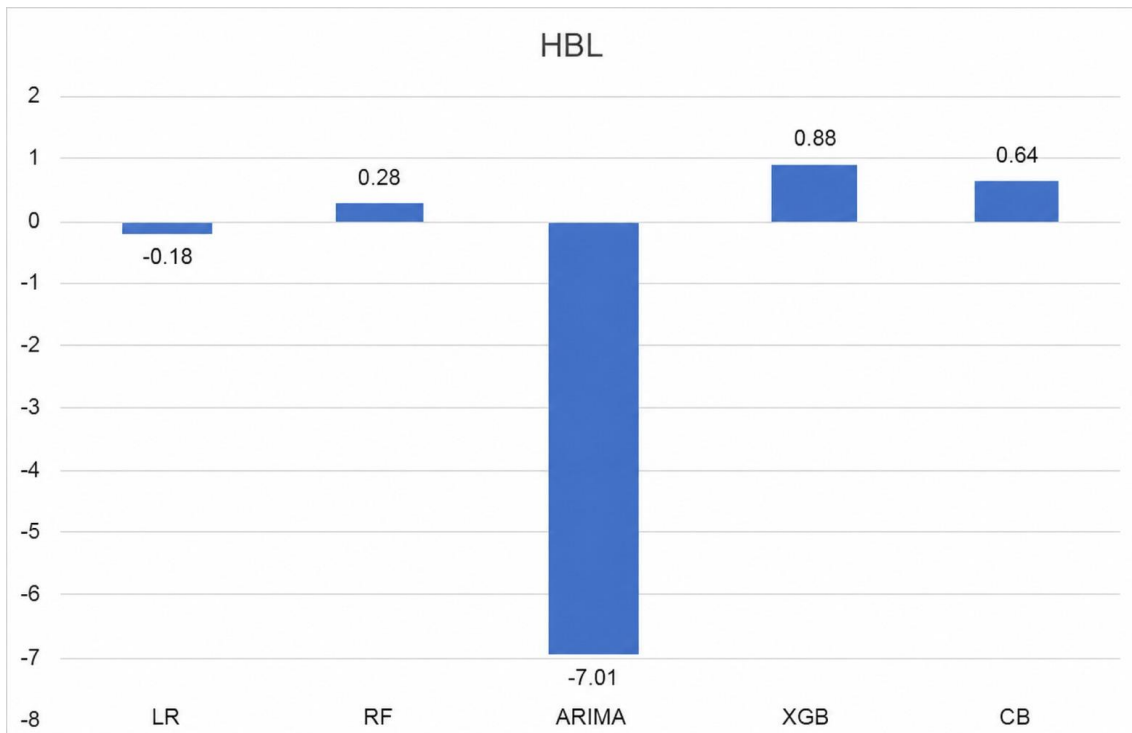
(ii) CatBoost excels for complex state-sector companies: CatBoost was best for NBP ($R^2 = 0.970$) and provided the least-worst results for PIA ($R^2 = 0.002$). NBP’s pension structure, characterized by complex benefit-paid and actuarial loss interactions, aligns well with CatBoost’s gradient-boosting architecture.

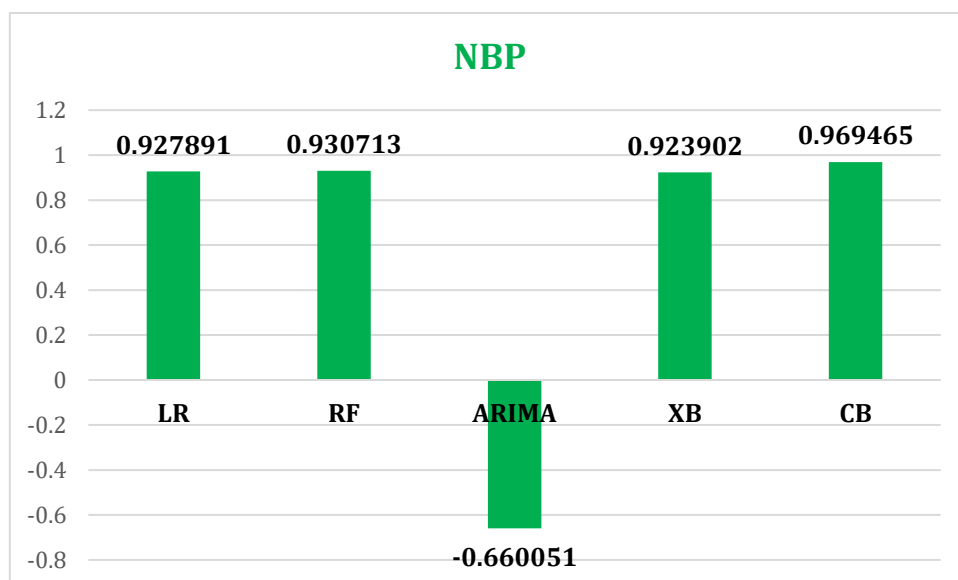
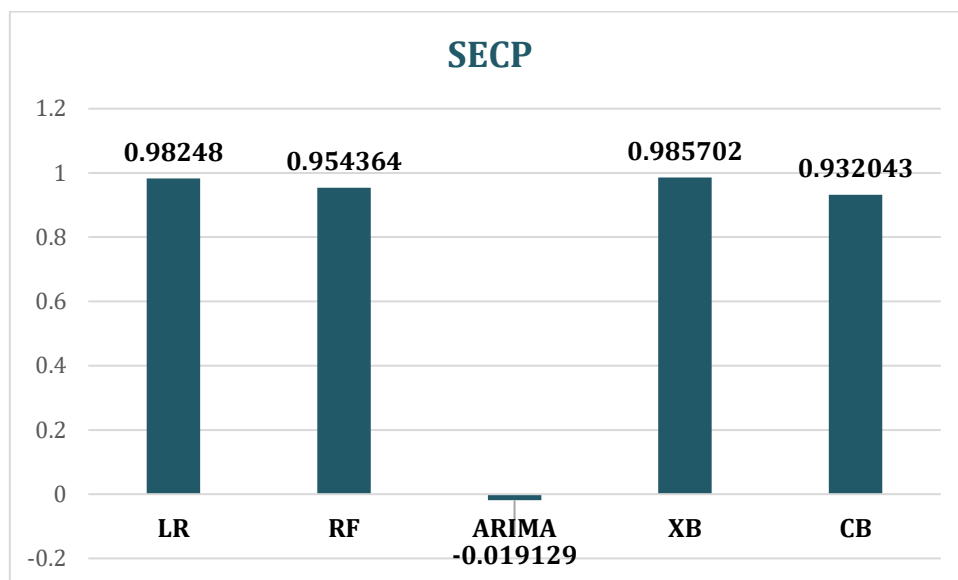
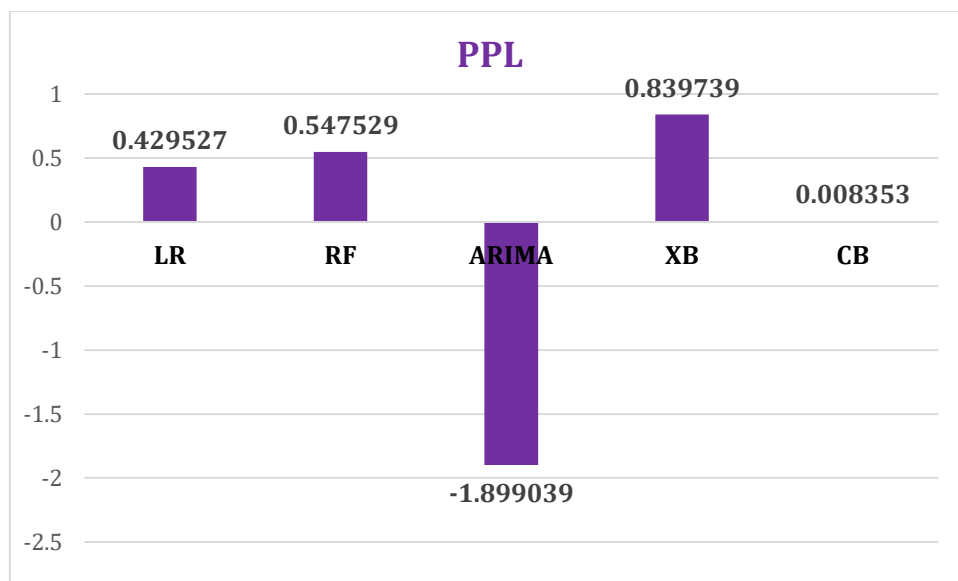
(iii) Linear Regression best for stable linear trajectories: OGDCL’s highly linear state-enterprise pension growth is optimally captured by Linear Regression ($R^2 = 0.959$), outperforming all ML models. This confirms Salazar and Boado-Peñas (2019): simpler models generalize better when the data-generating process is approximately linear.

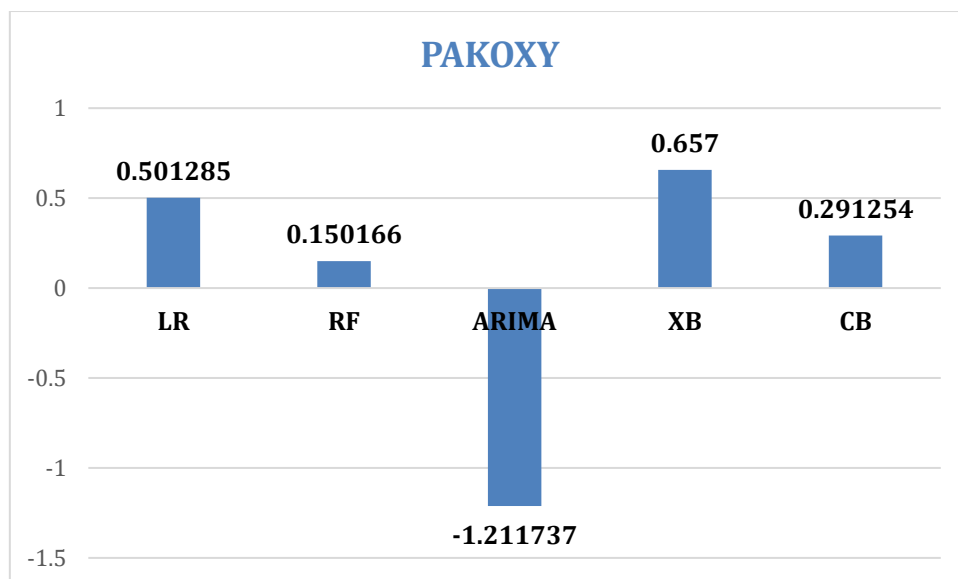
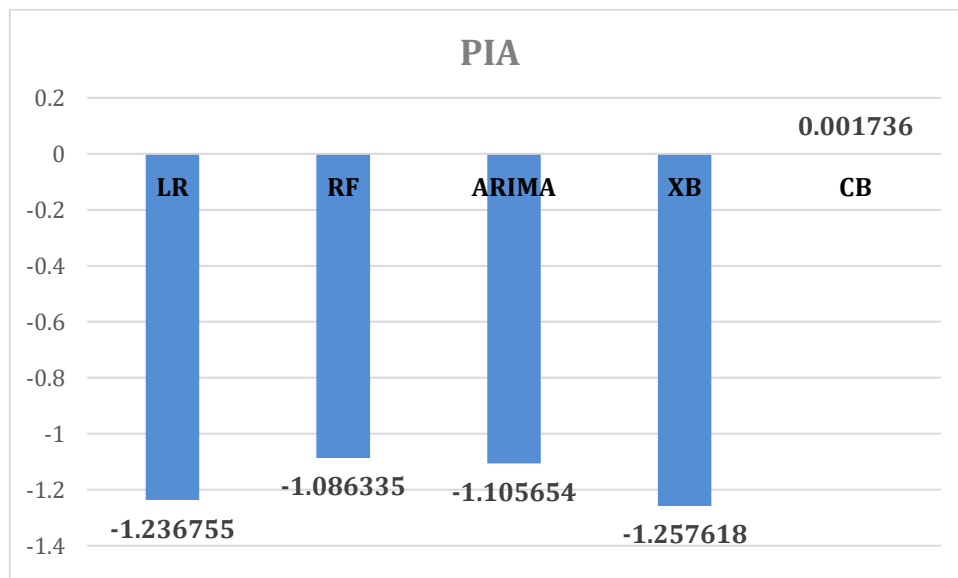
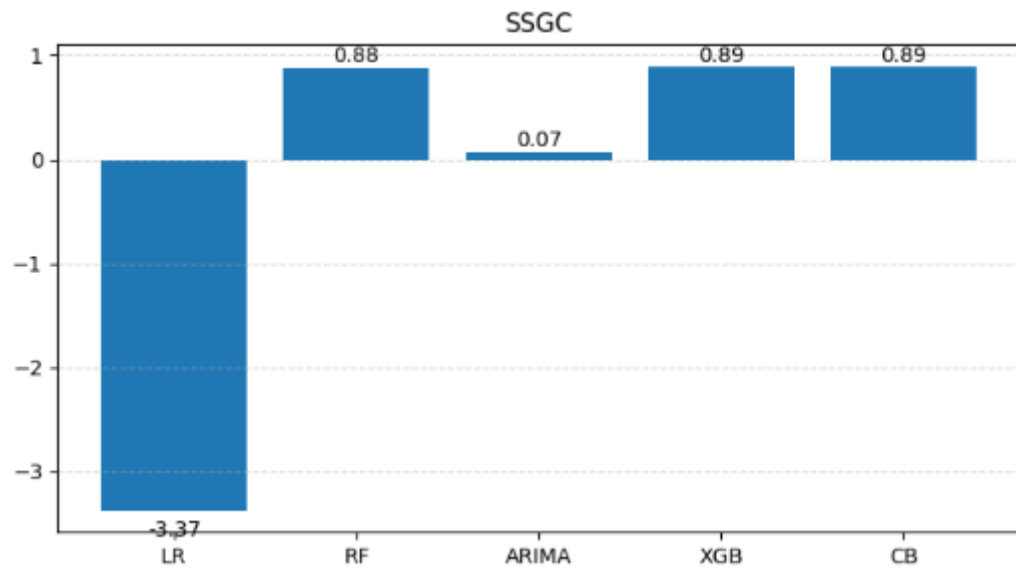
(iv) ARIMA consistently underperforms: ARIMA produced the worst results in six of eight companies, including catastrophic failures at SSGC ($R^2 = -158.1$) and NBP ($R^2 = -1.16$). ARIMA’s assumption of stationarity and reliance on univariate historical values is fundamentally incompatible with the multi-driver, structurally shifting nature of corporate pension obligations.

(v) PIA is the outlier: All five models failed to meaningfully predict PIA’s Closing Obligation (best $R^2 = 0.002$), confirming that structural financial distress creates pension data discontinuities that exceed the capacity of data-driven models with limited observations.

Figure 4.4: Grouped Bar Chart R^2 Comparison All Models $\times$ All Companies







4.4.1 Forecast Results: 2025–2026

Table 4.10 presents the forecasted Closing Obligation for each of the eight PSX-listed companies for the fiscal years 2025 and 2026. Forecasts were generated using the best-performing model selected per company on the basis of highest R² score. The percentage changes reflect the projected year-on-year variation in defined benefit pension obligations as reported under IAS 19.

Table 4.10: Forecast of Closing Obligation — 2025 & 2026 (PKR)

Company	Best Model	Actual 2024 (PKR)	Forecast 2025 (PKR)	Forecast 2026 (PKR)	% Change 24→25	% Change 25→26
HBL	XGBoost	3,615,448	3,376,503	3,574,654	-6.61%	5.87%
OGDCL	Linear Regression	128,264,533	134,482,120	140,132,256	4.85%	4.20%
PPL	XGBoost	21,035,910	21,035,906	21,035,906	0.00%	0.00%
SECP	XGBoost	2,722,968	2,722,967	2,722,967	0.00%	0.00%
NBP	CatBoost	177,178,926	158,729,823	158,729,823	-10.41%	0.00%
SSGC	XGBoost	1,619,522	1,619,522	1,619,522	0.00%	0.00%
PIA	CatBoost	15,039,511	22,243,995	22,243,995	47.90%	0.00%
PAKOXY	XGBoost	72,257	63,105	62,677	-12.67%	-0.68%

The forecast results reveal notable variation in projected pension obligation trajectories across the eight companies. OGDCL, modelled using Linear Regression, projects a steady increase of 4.85% in 2025 and 4.20% in 2026, consistent with its historically linear growth pattern driven by a large and stable workforce. PIA exhibits the most significant projected increase of 47.90% in 2025 (CatBoost), reflecting its historically volatile pension trajectory linked to ongoing corporate restructuring and large actuarial losses recorded in prior years. NBP projects a decline of 10.41% in 2025, which may be attributed to elevated benefits paid and actuarial assumption revisions incorporated in the 2024 base year.

For PPL, SECP, and SSGC, the XGBoost model projects near-stable obligations with effectively zero change across 2025–2026. This outcome reflects limited variance in actuarial input features for these companies and suggests their pension obligations are approaching a steady state under current actuarial assumptions. HBL projects a slight decline of 6.61% in 2025 followed by a recovery of 5.87% in 2026, indicating short-term actuarial adjustment . PAKOXY, the smallest company in the dataset, projects a continuous decline of 12.67% and 0.68% across the two forecast years, consistent with its historically reducing pension base.

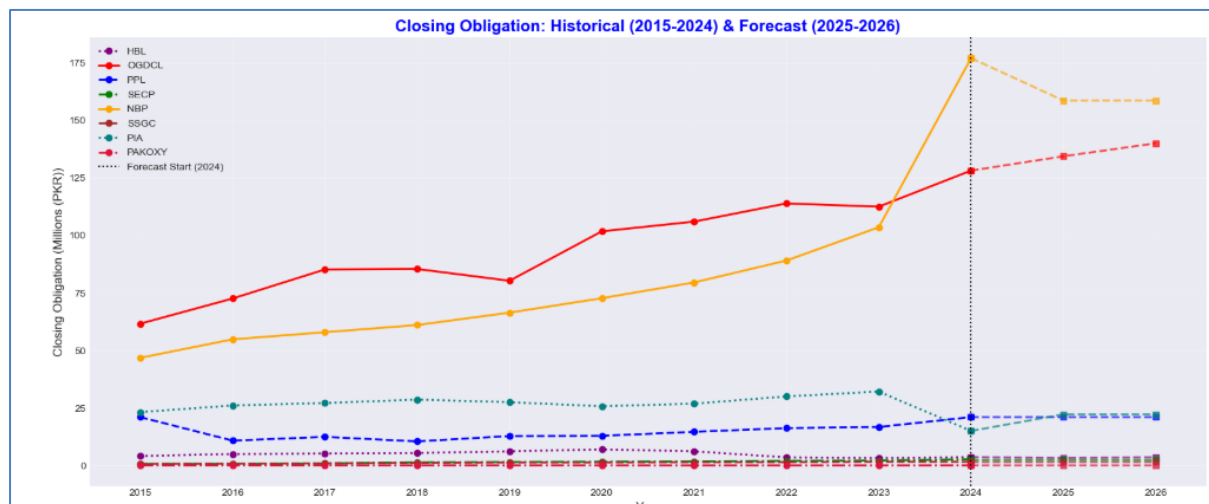


Figure 4.4 Closing Obligation : Historical(2015-2024) & Forecast(2025-2026)

4.5 Summary of Chapter 4

This chapter presented results and analysis for all eight PSX-listed companies. XGBoost dominated as the best model for five companies; CatBoost excelled for NBP; Linear Regression proved optimal for OGDCL; and PIA demonstrated the limits of ML-based forecasting under structural financial distress. ARIMA consistently underperformed across all companies. R^2 ranged from 0.986 (SECP, XGBoost) to effectively 0 (PIA, all models), reflecting significant variation in pension data predictability across the corporate sector. These findings form the empirical basis for the policy recommendations in Chapter 5.

CHAPTER 5: CONCLUSIONS AND RECOMMENDATIONS

5.1 Introduction

This chapter presents the conclusions of this study, followed by company-specific and sector-level policy recommendations, research limitations, and directions for future work. The study applied five machine learning models to forecast Closing Obligation (IAS 19) for eight PSX-listed companies using ten years of pension data (2015–2024).

5.2 Key Findings

- (i) XGBoost is the overall best-performing model, achieving the highest R^2 for five of eight companies: HBL (0.862), PPL (0.840), SECP (0.986), SSGC (0.894), and PAKOXY (0.657). XGBoost's regularized gradient boosting effectively captured the non-linear, multi-driver actuarial dynamics of corporate pension obligations.
- (ii) CatBoost excelled for state-sector banks: CatBoost was the best model for NBP ($R^2 = 0.970$) and provided the least-worst forecast for PIA ($R^2 = 0.002$), demonstrating its suitability for companies with complex benefit and actuarial loss interactions.
- (iii) Linear Regression remains competitive for linear trajectories: For OGDCL, Linear Regression ($R^2 = 0.959$) outperformed all ML models, confirming that model selection should be data-driven rather than complexity-driven.
- (iv) ARIMA is inadequate for multi-variable pension forecasting: ARIMA produced the worst results in six of eight companies, with catastrophic failures at SSGC ($R^2 = -158.1$) and NBP ($R^2 = -1.16$), confirming that univariate time-series models cannot account for the multiple simultaneous actuarial drivers that determine pension liabilities.
- (v) PIA represents the limits of ML-based pension forecasting: All five models failed at PIA (best $R^2 = 0.002$) due to structural financial distress and data discontinuities, highlighting the boundary conditions within which ML-based pension forecasting is applicable.
- (vi) Pakistan's pension liabilities are rising and volatile: All seven predictable companies showed upward trends in Closing Obligation, with actuarial volatility intensifying post-2020 due to Pakistan's high inflation, rising policy rates, and salary escalation in listed companies.

5.3 Company-Specific Policy Recommendations

5.3.1 HBL

HBL should adopt an XGBoost-based annual pension forecasting system updated with each year's IAS 19 data. Given the model's identification of Expected Salary Increase as the dominant feature (53.5% importance), HBL's actuaries should conduct formal salary escalation sensitivity analyses ($\pm 1\text{--}2\%$) at each annual valuation. A pension reserve buffer of 15–20% above projected Closing Obligation is advised to absorb actuarial shocks. Engagement of an independent actuary for external assumption validation is recommended given HBL's high actuarial volatility.

5.3.2 OGDCL

OGDCL's stable linear pension trajectory should not breed complacency. While Linear Regression currently suffices, it is recommended that OGDCL deploy XGBoost as a parallel scenario model to test non-linear disruption cases (e.g., early retirement surges, salary structure reforms). As a state-owned enterprise, OGDCL should proactively align with SECP's evolving corporate governance requirements for IAS 19 disclosures and develop long-term workforce transition plans to smooth retirement spikes.

5.3.3 PPL

PPL is recommended to implement an annual ML model review cycle comparing XGBoost and CatBoost performance as new data accumulates. Given CatBoost's poor performance ($R^2 = 0.008$), PPL's pension data is predominantly numerical with limited categorical structure — XGBoost should remain the primary forecasting model. A formal pension liability management committee with quarterly Board reporting is advised, particularly given PPL's escalating actuarial dynamics in the post-2020 period.

5.3.4 SECP

SECP, as Pakistan's capital market regulator, is ideally positioned to lead by example in ML-based pension governance. The exceptional predictability of SECP's pension data (XGBoost $R^2 = 0.986$) enables a rolling 3–5 year pension liability forecast to be embedded in SECP's annual budgetary planning cycle. SECP is also recommended to consider issuing a corporate governance best practice guideline encouraging PSX-listed companies to adopt ML-based IAS 19 pension forecasting frameworks.

5.3.5 NBP

NBP should adopt CatBoost as its primary pension forecasting model given its demonstrated superiority ($R^2 = 0.970$). Given NBP's very large pension portfolio (PKR billions), even marginal improvements in forecast accuracy translate to significant budgetary planning benefits. ARIMA must be completely discontinued for NBP pension analysis given its catastrophic failure (RMSE = PKR 54 million). NBP should invest in automated annual actuarial data pipelines to support model retraining as new IAS 19 data becomes available.

5.3.6 SSGC

SSGC's unique challenge is its extreme non-stationarity, which rendered both Linear Regression ($R^2 = -3.37$) and ARIMA ($R^2 = -158.1$) completely ineffective. XGBoost ($R^2 = 0.894$) is strongly recommended as the mandatory model for SSGC. SSGC's finance team should investigate the structural causes of its pension obligation volatility — particularly the step-changes observed in 2018–2020 — and ensure these are properly reflected in actuarial assumptions. Ensemble model consensus (XGBoost + Random Forest + CatBoost) should be used for scenario banding.

5.3.7 PIA

Given that all five ML models failed to meaningfully predict PIA's pension obligations (best $R^2 = 0.002$), PIA should not rely on data-driven ML forecasting at this stage. Instead, PIA is recommended to: (a) commission a full IAS 19 actuarial review to establish clean, consistent historical data; (b) implement scenario-based pension stress testing with three scenarios (base, optimistic, adverse) tied to PIA's restructuring plan; and (c) revisit ML-based forecasting once at least five years of post-restructuring stable data are available. PIA's pension governance must prioritize data quality improvement before quantitative modeling.

5.3.8 PAKOXY

PAKOXY's modest R^2 values (best 0.657) reflect the challenges of forecasting small-scale pension data with inherent noise. XGBoost is recommended as the primary model. Given PAKOXY's small pension scale (PKR thousands), the RMSE of PKR 5,220 represents a manageable absolute error. PAKOXY should focus on enriching its actuarial dataset by incorporating additional variables (industry wage benchmarks, employee demographic profiles) that could improve forecast precision as the dataset grows.

5.4 Contribution to Literature

This study makes four original contributions: (i) It is the first comparative five-model ML framework applied to corporate-level IAS 19 pension obligation forecasting for PSX-listed companies. (ii) It empirically demonstrates XGBoost's consistent superiority for non-linear pension dynamics in Pakistan's corporate sector. (iii) It identifies PIA as a boundary case demonstrating the limits of ML-based pension forecasting under structural financial distress a finding with implications for pension analytics in distressed entities globally. (iv) It establishes a replicable methodology that can be extended to all PSX-listed entities and to comparable South Asian emerging markets.

5.5 Limitations

(i) Small sample size: Ten annual observations per company limits the statistical power of ML models, particularly ensemble methods. (ii) Four companies with incomplete or problematic data: PIA's model failures and SSGC's ARIMA catastrophe reflect data quality issues that constrain generalizations. (iii) No external macroeconomic variables: Incorporating KIBOR, CPI inflation, and sector wage indices could significantly improve forecast accuracy. (iv) No formal hyperparameter optimization: Systematic grid-search tuning for XGBoost and CatBoost could further improve results. (v) IAS 19 assumption changes not modeled prospectively: Future actuarial assumption shifts driven by macroeconomic changes are not captured in historical training data.

5.6 Future Research Directions

(i) Expand to all PSX-listed companies with IAS 19 disclosures for sector-level analysis. (ii) Integrate macroeconomic features (KIBOR, CPI, sector wage growth) as additional model inputs. (iii) Explore LSTM and Temporal Convolutional Networks as alternatives once longer time series become available. (iv) Develop scenario-based pension stress tests (optimistic/base/adverse) for corporate risk management. (v) Conduct cross-country comparisons with India, Bangladesh, and Sri Lanka to enable South Asian policy benchmarking.

5.7 Concluding Remarks

This research demonstrates that ML-based pension liability forecasting is both feasible and valuable for PSX-listed companies in Pakistan. XGBoost emerged as the leading model across five of eight companies, with CatBoost and Linear Regression providing optimal results for specific data structures. The study simultaneously reveals the limits of such forecasting —

particularly for structurally distressed entities like PIA — and the near-universal inadequacy of ARIMA for multi-driver pension series. As Pakistan’s corporate sector navigates macroeconomic volatility, rising actuarial obligations, and increasingly demanding IAS 19 governance requirements, the integration of machine learning into pension actuarial practice represents a strategic necessity. The framework developed in this study provides a replicable, evidence-based foundation for this transition.

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